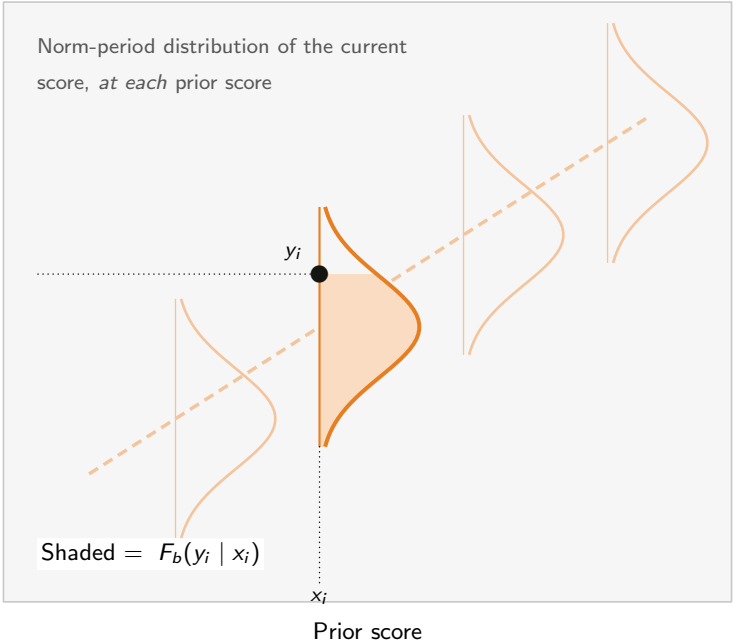


A. Condition on the prior score

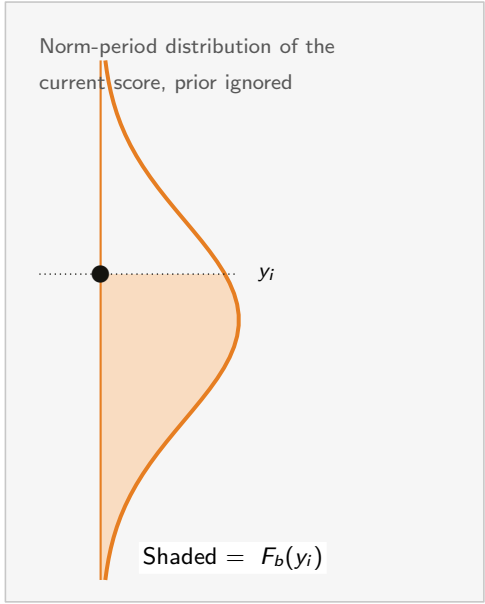
This is the baseline SGP



average over
the prior
 $F_b(y) = E_X[F_b(y | X)]$

B. Drop the conditioning

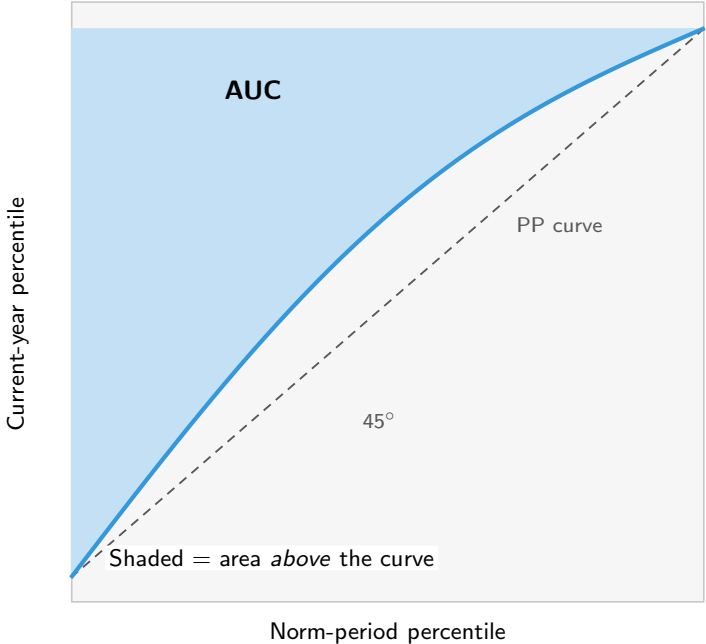
One marginal distribution



average over
the cohort
 $E[F_b(Y_c)]$

C. Average over the cohort

Ho's comparison: a probability-probability (PP) plot



$E[F_b(Y_c)] = \Pr(Y_b < Y_c) = \text{AUC}$

$V = \sqrt{2} \Phi^{-1}(\text{AUC})$

Notation. x_i, y_i : one student's prior and current score. F_b : the norm-period distribution function, so $F_b(y)$ is the share of norm-period students scoring below y ; $F_b(y | x)$ its conditional form, given prior score x .

Y_b, Y_c : scores drawn at random from the norm period and from the current year. $E[\cdot]$: an average. Φ^{-1} : the inverse standard normal distribution function.

A conditions on the prior. B and C do not.